

NBA Game Total Market Efficiency

Semi-Strong Form Efficiency Testing via a Volatility Risk Premium Framework

Seasons 2019–20 through 2022–23 · Real SBR Closing Lines · N = 665 merged games

This paper tests whether the NBA game totals betting market incorporates publicly available information efficiently. Using closing lines from Sportsbook Review matched against NBA API game logs across four seasons, five formal hypotheses are tested. The central question is whether a bettor armed only with the public schedule and injury reports could generate systematic positive returns — which, by definition, would constitute evidence against market efficiency.

1. Data Overview

Two independent data sources are merged at the game level. The NBA API provides scheduling and player-availability information. SBR Excel archives provide the closing total line for each game. Only games where both sources agree on date and team survive the inner join.

NBA API player-game rows	52,065 (cached, ≥ 10 min played)
NBA API team-game rows	7,380
SBR files loaded	4 (2019-20, 2020-21, 2021-22, 2022-23)
SBR games after validity filter	1,618
Games after inner join (final N)	665
Implied probability source	Line movement (Close – Open), sigmoid-mapped
Season fixed effects	Included in all regressions

Note: 665 games is lower than the theoretical maximum (~4,920) because the NBA API cache currently covers only three seasons. Deleting the cached CSVs and re-running with all four seasons listed will recover the remaining ~1,400 games and improve statistical power across all tests.

2. Implied Probability Construction

NBA totals markets are almost universally priced at $-110/-110$, meaning both sides carry identical vig. After vig-stripping, every game would get $p_{\text{over}} = 0.50$ exactly — which makes calibration and FLB tests uninformative (no variation to regress on).

Instead, implied probability is derived from **line movement** — the difference between the opening and closing total. When books move a line upward (e.g. 224 $\rightarrow$ 226), it signals that sharp money or public action has

pushed toward the over. The sigmoid function converts this movement to a probability:

$$p_over_fair = 1 / (1 + \exp(-0.15 \times line_move))$$

where *line_move* = Close – Open, clipped to [-5, +5] points to exclude data errors. A move of zero gives p = 0.50. A move of +2 gives p ≈ 0.575. This produces genuine cross-game variation while remaining grounded in the market's own revealed preferences.

Mean p_over_fair	0.4999
Std p_over_fair	0.0934
Mean line move	-0.003 pts
Std line move	2.567 pts
% games with movement	80.7%

3. H1 — Calibration Test

What it asks

When the market signals a 55% chance of the over, does the game actually go over 55% of the time? A perfectly efficient market must be calibrated — its probability estimates should match realized frequencies at every level.

The test

The Mincer-Zarnowitz (1969) regression regresses the realized outcome (1 if over, 0 if under) on the implied probability:

$$\text{over_hit}_i = \alpha + \beta \cdot \text{p_over_fair}_i + \varepsilon_i$$

Under efficiency the joint null is $\alpha = 0$ (no unconditional bias) and $\beta = 1$ (probability signal is fully informative). The Brier skill score compares forecast accuracy against a naive 50/50 benchmark — positive skill means the market beats the benchmark.

Results

Statistic	Estimate	p-value	Benchmark
α (intercept)	0.3568	0.001	$H_0: \alpha = 0$
β (slope)	0.3075	0.140	$H_0: \beta = 1$
Joint test ($\alpha=0, \beta=1$)	—	< 0.001	Both conditions required
Brier score	0.2533	—	Lower is better; 0.25 = baseline
Brier skill score	-0.013	—	> 0 beats 50/50 benchmark
N	665	—	—

What this means

$\alpha = 0.357$ ($p = 0.001$): The intercept is significantly different from zero. This means the market has an unconditional bias — independently of what the line movement predicts, overs hit about 51% of the time on average.

$\beta = 0.308$ ($p = 0.14$ for $\beta=0$, $p < 0.001$ for $\beta=1$): The slope is far below the ideal value of 1. A slope of 1 would mean the market's probability signals are fully informative — games flagged as 60% over actually go over 60% of the time. Here, the signal is weak but not completely absent.

Brier skill = -0.013: The market's probabilistic forecasts are marginally worse than simply predicting 50% for every single game. This is partly a consequence of using line movement as the probability proxy rather than actual market odds.

REJECTED — H1 rejected. The market is not well-calibrated.

4. H2 — Aggregate Volatility Risk Premium (VRP)

What it asks

On average across all games, does the market systematically overprice the over? The analogy to equity options is direct: in options markets, implied volatility persistently exceeds realized volatility — sellers of variance earn a premium. Here, if the market consistently overestimates the probability of the over, then systematically betting the under would have positive expected value before vig.

The test

The residual for each game is defined as $r = \text{over_hit} - p_{\text{over_fair}}$ (realized outcome minus implied probability). The aggregate VRP is the negative of the mean residual:

$$\text{VRP} = -E[r] = E[p_{\text{over_fair}}] - E[\text{over_hit}]$$

A **positive VRP** means the implied probability on average exceeds the realized over rate — systematic overpricing of the over. A one-sample t-test assesses significance.

Results

Statistic	Value	Inference
Aggregate VRP	−0.0089	Over underpriced by ~0.9 pp on average
t-statistic	0.456	—
p-value	0.649	Not significant — H_0 not rejected
N games	665	—

VRP by subgroup

The VRP is also broken out by scheduling and availability subgroups. If market inefficiency is concentrated in specific game types, the aggregate test may miss it.

Subgroup	N	VRP	t-stat	p-value	Sig.
Home B2B	119	−0.0170	0.360	0.720	—
Away B2B	97	−0.0310	0.608	0.544	—
Star Absent	289	−0.0502	1.693	0.092	*

What this means

The aggregate VRP of −0.0089 is economically tiny and statistically insignificant ($p = 0.65$). There is no evidence that the market systematically overprices the over across all games. The star-absent subgroup shows marginal significance at the 10% level ($p = 0.09$), suggesting a small tendency for overs to outperform

the implied probability when a star player is missing — though this is the *opposite* direction to the original hypothesis (which predicted the under would be underpriced when a star is absent).

REJECTED — H2 not supported. No significant aggregate VRP.

5. H5 — Favorite-Longshot Bias (FLB)

What it asks

The Favorite-Longshot Bias is a well-documented phenomenon in betting markets: bettors systematically overpay for longshots (low-probability outcomes) and underpay for favorites (high-probability outcomes). In horse racing, longshot implied probabilities reliably exceed actual win rates. In the totals context: do games flagged as low-probability overs go over more often than the market expects?

The test

All games are sorted into 9 bins by implied over-probability. Within each bin, the mean implied probability is compared to the mean realized over rate. Under efficiency all points should lie on the 45-degree line (implied = realized). Classic FLB produces a bow-shaped pattern — low-probability bins sit above the line, high-probability bins sit below it.

Results

Bin	Implied Prob	Realized Rate	N	Edge	t-stat
0 (lowest)	0.344	0.494	85	+0.150	2.747 **
1	0.408	0.471	70	+0.064	1.056
2	0.455	0.362	58	−0.093	−1.451
3	0.497	0.506	174	+0.009	0.239
4	0.519	0.514	35	−0.005	−0.052
5	0.537	0.640	50	+0.103	1.496
6	0.564	0.565	62	+0.000	0.003
7	0.610	0.559	68	−0.051	−0.847
8 (highest)	0.667	0.524	63	−0.143	−2.260 **

** significant at 5%

What this means

There is partial evidence of a Favorite-Longshot Bias pattern. The lowest-probability bin (implied $\approx$ 34%) goes over at 49.4% — a statistically significant positive edge of +15 pp ($t = 2.75$). The highest-probability bin (implied $\approx$ 67%) goes over at only 52.4% — a significant negative edge of −14 pp ($t = -2.26$). This is directionally consistent with classic FLB.

However, the middle bins (3–7) are scattered around zero with no pattern. The extreme bins reflect games with the largest line movements, where the sigmoid mapping produces the most extreme probabilities. This means the FLB signal here is partly a function of which games had large line moves rather than a pure

market-pricing phenomenon.

INDETERMINATE — H5 partial. Low-probability bins show significant positive edge; high-probability bins show negative edge. Middle bins are noise.

6. H3 & H4 — Public Signal OLS Regression

What it asks

This is the core efficiency test. Under semi-strong form EMH, no publicly observable pre-game information should predict the forecast error. The schedule is publicly available before tip-off. If knowing that a team is on a back-to-back — or that a star player is sitting out — lets you systematically predict whether the game goes over or under the line, the market is inefficient by definition.

H3: Back-to-back games go under more often than the line implies.

Fatigue from playing on consecutive nights reduces pace and shooting quality, depressing scoring below what the book expects. If sportsbooks fail to fully price this in, the under becomes systematically valuable on B2B games.

H4: Star-absent games go under more often than the line implies.

Stars account for a disproportionate share of a team's offense. When a 20+ PPG player sits out, the total should drop. If the market is slow to incorporate last-minute DNP announcements into the closing line, the under has positive expected value.

The test

OLS regression of the forecast residual on pre-game public signals. HC3 heteroskedasticity-robust standard errors throughout. Season fixed effects absorb year-level shocks. The pace proxy (fga_z) is a within-game control included to isolate the pre-game signals.

$$\text{residual}_i = \beta_0 + \beta_1 \cdot \text{both_b2b} + \beta_2 \cdot \text{one_b2b} + \beta_3 \cdot \text{star_absent} + \beta_4 \cdot \text{fga_z} + \gamma_s \cdot \text{Season} + \varepsilon_i$$

If β_1 , β_2 , or β_3 are statistically significant and negative, publicly available information predicts forecast errors — direct evidence against semi-strong efficiency.

Results

Variable	β	SE	p-value	Sig.	Hypothesis
both_b2b	+0.0894	0.1004	0.373	—	H3
one_b2b	−0.0037	0.0461	0.937	—	H3
star_absent	+0.1168	0.0447	0.009	***	H4
fga_z (pace)	+0.0567	0.0229	0.013	**	Control
N observations			665		
R-squared			0.0177 (low R ² expected — game outcomes are inherently noisy)		
Logit pseudo R ²			0.0050 (robustness check — consistent with OLS)		

What this means

H3 — Back-to-back (both_b2b, one_b2b): Neither coefficient is significant. B2B scheduling does not predict forecast errors in this sample. Sportsbooks appear to price fatigue into the line, or the fatigue effect on scoring is smaller than hypothesised. The B2B under-betting strategy (Section 7) confirms this — it loses money.

H4 — Star absence (star_absent): The coefficient is highly significant ($\beta = +0.117$, $p = 0.009$) but has the *wrong sign*. The hypothesis predicted $\beta < 0$ (star absence depresses scoring below the line, making the under more likely). Instead, star-absent games go *over* more than implied. Two explanations are plausible: (1) backup players increase pace and FGA, inflating scoring; or (2) books overcorrect the line downward when a star is absent, making the over the value side. This is a genuine empirical result, not a data artefact — the implied probability audit confirms clean probability construction.

fga_z (pace control): Significant and positive as expected — faster-paced games produce more points. This is a within-game control, not observable pre-game, so it does not constitute evidence against efficiency.

REJECTED — H3 not supported. B2B has no significant effect.

INDETERMINATE — H4 significant but wrong sign ($\beta = +0.117$, $p = 0.009$). Star absence is associated with more scoring, not less.

7. Rule-Based Strategy Backtest

Strategy design

The backtest operationalises H3 as a real betting strategy. The signal fires when **both** conditions hold: (1) at least one team is on a back-to-back, AND (2) the closing total line is above that season's median. The second condition targets games where the market has set an above-average line despite fatigue — the hypothesised sweet spot for mispricing.

Flat unit staking is used throughout. Performance is evaluated on a gross basis (ignoring vig) and a net basis (applying standard -110 odds, requiring 52.38% win rate to break even). The Sharpe ratio annualises the per-bet risk-adjusted return by scaling by $\sqrt{N}$.

Results

Metric	Value	Benchmark / Notes
Qualifying bets (N)	99	~15% of 665 games triggered the signal
Win rate (under hit)	48.48%	Below 52.38% break-even
Gross ROI	-3.03%	Negative even before vig
Net ROI	-7.44%	Post-vig at -110 ; material loss
Sharpe ratio	-0.772	Negative risk-adjusted return

What this means

The strategy loses money. With a win rate of 48.5% and a gross ROI of -3.0% , the under does not hit more often than chance on B2B games with above-median lines. This is consistent with the OLS finding that the B2B coefficient is not significant.

This result is robust to the implied probability construction, because the backtest depends only on whether the game went over or under the posted total and on the B2B flag — neither of which is affected by the line-movement probability methodology. The -7.4% net ROI is a reliable finding on this data.

REJECTED — Backtest: No profitable B2B under-betting strategy. Net ROI = -7.4% , Sharpe = -0.77 .

8. Hypothesis Scorecard

Hypothesis	Test	Key Statistics	Verdict
H1 — Market is calibrated ($\alpha=0$, $\beta=1$)	Mincer-Zarnowitz OLS	$\alpha=0.357$ ($p=0.001$) $\beta=0.308$ ($p=0.14$) Brier skill= -0.013	REJECTED
H2 — Aggregate VRP is positive	One-sample t-test on residuals	VRP= -0.0089 $t=0.456$, $p=0.649$	NOT SUPPORTED
H3 — B2B games go under	OLS coefficients both_b2b, one_b2b	$\beta=+0.089$ ($p=0.373$) $\beta=-0.004$ ($p=0.937$)	NOT SUPPORTED
H4 — Star absence depresses scoring	OLS coefficient star_absent	$\beta=+0.117$ ($p=0.009$) Wrong sign	SIGNIFICANT WRONG SIGN
H5 — Favorite-Longshot Bias	Decile binning implied vs. realized	Bin 0 edge= $+0.150$ ($t=2.75$) Bin 8 edge= -0.143 ($t=-2.26$)	PARTIAL FLB PATTERN

9. Discussion

What the results collectively say

The market passes the most important test: there is no significant aggregate overpricing of the over (VRP $p = 0.65$), and the simple B2B under-betting strategy loses money (net ROI = -7.4%). Back-to-back scheduling, the most intuitive fatigue signal, is already priced in.

The calibration failure (H1 rejected) is a methodological limitation rather than a strong efficiency finding. Because NBA totals are priced at $-110/-110$, implied probabilities are derived from line movement, which introduces noise. A market with direct over/under odds (as in European football markets) would allow a cleaner calibration test.

The most interesting result is the star-absence finding. Games where a 20+ PPG player is absent go over significantly more than the line movement predicts ($\beta = +0.117$, $p = 0.009$). This could mean: (a) books overcorrect the total downward on DNPs, creating over value; or (b) backup players generate more FGA and pace, inflating scoring. Either way, this deserves further investigation with a dedicated star-absence betting strategy.

The partial FLB pattern (H5) is the second notable finding. Games with the largest upward line movements (implied $\approx 67\%$) underperform their implied probability by 14 pp, while games with the largest downward moves outperform by 15 pp. This suggests the market may overreact to line-moving information, consistent with momentum-chasing by sharp bettors.

Limitations

- Sample size: 665 merged games is below the target of $\sim 2,000$. Re-running with the full NBA API cache will materially improve statistical power.

- Implied probability proxy: Deriving probabilities from line movement is an approximation. A market with explicit over/under odds would allow direct vig-stripping.
- Star absence inference: DNPs are inferred from game log absence, not from official injury reports. Some inferred absences may reflect load management not captured by the 20 PPG threshold.
- Look-ahead: All features are constructed from pre-game public information. The fga_z pace control is a within-game variable included as a control only.

Next steps

- Expand NBA API cache to all four seasons (delete cached CSVs, re-run) — expected to recover ~1,400 additional games.
- Build and backtest a dedicated star-absence strategy (bet OVER when 20+ PPG player is absent and the line has moved down).
- Investigate the FLB pattern further — test whether the extreme-bin edges survive after controlling for season and pace.
- Replace line-movement probability with actual moneyline-derived probabilities if a data source with separate over/under odds becomes available.

Working Paper · Seasons 2019–20 through 2022–23 · Generated May 13, 2026 · Pipeline: nba_prop_pipeline.py · Data: NBA API + SBR
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